

Homework 5

Andrew Hwang

Math 23

Haohua Deng

May 12, 2026

1 7.4

Question 3. Given the system

$$\mathbf{x}' = \begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}^{(1)} = \begin{pmatrix} 5 \cos t \\ 2 \cos t + \sin t \end{pmatrix}, \quad \mathbf{x}^{(2)} = \begin{pmatrix} 5 \sin t \\ -\cos t + 2 \sin t \end{pmatrix}.$$

(c) Show that the given functions form a fundamental set of solutions of the given system.

(d) Find the solution of the given system that satisfies the initial condition $\mathbf{x}(0) = (1, 2)^T$.

(e) Find $W[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}](t)$.

Solution. For part (c), suppose $\mathbf{x}^{(1)} = k\mathbf{x}^{(2)}$ for some constant k . Then:

$$5 \cos t = 5k \sin t \implies k = \cot t \quad \text{and} \quad 2 \cos t + \sin t = k(-\cos t + 2 \sin t).$$

Since $k = \cot t$ depends on t and is not a constant, this is a contradiction. Therefore $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ are linearly independent and form a fundamental set of solutions. For part (d),

$$\mathbf{x} = \mathbf{\Psi}(t)\mathbf{c} = \begin{pmatrix} 5 \cos t & 5 \sin t \\ 2 \cos t + \sin t & -\cos t + 2 \sin t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

Applying $\mathbf{x}(0) = (1, 2)^T$ gives

$$\begin{pmatrix} 5 & 0 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

From the first equation: $c_1 = \frac{1}{5}$. Substituting into the second: $\frac{2}{5} - c_2 = 2 \implies c_2 = -\frac{8}{5}$. Therefore,

$$\mathbf{x} = \frac{1}{5}\mathbf{x}^{(1)} - \frac{8}{5}\mathbf{x}^{(2)} = \frac{1}{5} \begin{pmatrix} 5 \cos t - 8 \cdot 5 \sin t \\ (2 \cos t + \sin t) - 8(-\cos t + 2 \sin t) \end{pmatrix} = \begin{pmatrix} \cos t - 8 \sin t \\ 2 \cos t - 3 \sin t \end{pmatrix}.$$

For part (e), we compute the Wronskian directly:

$$\begin{aligned} W[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}](t) &= \begin{vmatrix} 5 \cos t & 5 \sin t \\ 2 \cos t + \sin t & -\cos t + 2 \sin t \end{vmatrix} \\ &= 5 \cos t(-\cos t + 2 \sin t) - 5 \sin t(2 \cos t + \sin t) \\ &= 5(-\cos^2 t + 2 \sin t \cos t - 2 \sin t \cos t - \sin^2 t) = \boxed{-5}. \end{aligned}$$

□

Question 12. Consider the vectors $\mathbf{x}^{(1)}(t) = \begin{pmatrix} t \\ 1 \end{pmatrix}$ and $\mathbf{x}^{(2)}(t) = \begin{pmatrix} t^2 \\ 2t \end{pmatrix}$.

- (a) Compute the Wronskian of $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$.
- (b) In what intervals are $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ linearly independent?
- (c) What conclusion can be drawn about the coefficients in the system of homogeneous differential equations satisfied by $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$?
- (d) Find this system of equations and verify the conclusions of part (c).

Solution. For part (a), we compute the Wronskian directly:

$$W[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}](t) = \begin{vmatrix} t & t^2 \\ 1 & 2t \end{vmatrix} = 2t^2 - t^2 = \boxed{t^2}.$$

For part (b), since $W = t^2 = 0$ only when $t = 0$, the vectors are linearly independent on any interval not containing $t = 0$. Therefore $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ are linearly independent on

$$\boxed{(-\infty, 0) \quad \text{and} \quad (0, \infty)}.$$

For part (c), by Abel's Formula,

$$W' = (p_{11}(t) + p_{22}(t))W.$$

Substituting gives

$$2t = (p_{11}(t) + p_{22}(t)) \cdot t^2 \implies p_{11}(t) + p_{22}(t) = \frac{2}{t}.$$

Since $\frac{2}{t}$ is discontinuous at $t = 0$, we conclude that at least one of the coefficient functions $p_{ij}(t)$ must be discontinuous at $t = 0$. For part (d), we find the system $\mathbf{x}' = P(t)\mathbf{x}$ satisfied by $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$. We need:

$$\begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \begin{pmatrix} t \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \begin{pmatrix} t^2 \\ 2t \end{pmatrix} = \begin{pmatrix} 2t \\ 2 \end{pmatrix}.$$

From the first system,

$$p_{11}t + p_{12} = 1, \quad p_{21}t + p_{22} = 0.$$

From the second system,

$$p_{11}t^2 + 2p_{12}t = 2t, \quad p_{21}t^2 + 2p_{22}t = 2.$$

Subtracting the first equation of both systems gives $p_{12} = 1$ and $p_{11} = 0$. Similarly, subtracting the second equations gives $p_{22} = \frac{2}{t}$ and $p_{21} = -\frac{2}{t^2}$. Therefore the system is

$$\boxed{\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -\frac{2}{t^2} & \frac{2}{t} \end{pmatrix} \mathbf{x}.$$

□

2 7.5

Question 7. Find the general solution of the given system of equations.

$$\mathbf{x}' = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix} \mathbf{x}$$

Solution. The characteristic equation is

$$\det \begin{pmatrix} 1-\lambda & 1 & 2 \\ 1 & 2-\lambda & 1 \\ 2 & 1 & 1-\lambda \end{pmatrix} = 0.$$

Expanding the determinant and simplifying gives

$$-\lambda^3 + 4\lambda^2 + \lambda - 4 = 0 \implies \lambda^3 - 4\lambda^2 - \lambda + 4 = 0.$$

Factoring by grouping:

$$\lambda^2(\lambda - 4) - (\lambda - 4) = 0 \implies (\lambda^2 - 1)(\lambda - 4) = 0 \implies \lambda_1 = -1, \quad \lambda_2 = 1, \quad \lambda_3 = 4.$$

For $\lambda_1 = -1$,

$$\begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(1)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 1$,

$$\begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(2)} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}.$$

For $\lambda_3 = 4$,

$$\begin{pmatrix} -3 & 1 & 2 \\ 1 & -2 & 1 \\ 2 & 1 & -3 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(3)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Therefore, the general solution is

$$\boxed{\mathbf{x} = c_1 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} e^t + c_3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{4t}.$$

□

Question 11. Solve the given initial value problem. Describe the behavior of the solution as $t \rightarrow \infty$.

$$\mathbf{x}' = \begin{pmatrix} -2 & 1 \\ -5 & 4 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Solution. The characteristic equation is

$$\det(A - \lambda I) = \det \begin{pmatrix} -2 - \lambda & 1 \\ -5 & 4 - \lambda \end{pmatrix} = (-2 - \lambda)(4 - \lambda) + 5 = \lambda^2 - 2\lambda - 3 = 0 \implies (\lambda - 3)(\lambda + 1) = 0.$$

So $\lambda_1 = -1$ and $\lambda_2 = 3$. For $\lambda_1 = -1$,

$$\begin{pmatrix} -1 & 1 \\ -5 & 5 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \implies \mathbf{x}_1 = \begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}.$$

For $\lambda_2 = 3$,

$$\begin{pmatrix} -5 & 1 \\ -5 & 1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & -\frac{1}{5} \\ 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(2)} = \begin{pmatrix} 1 \\ 5 \end{pmatrix} \implies \mathbf{x}_2 = \begin{pmatrix} e^{3t} \\ 5e^{3t} \end{pmatrix}.$$

Now, the fundamental matrix is

$$\Psi(t) = (\mathbf{x}_1 \quad \mathbf{x}_2) = \begin{pmatrix} e^{-t} & e^{3t} \\ e^{-t} & 5e^{3t} \end{pmatrix}.$$

At $t = 0$,

$$\Psi(0) = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix} \implies \Psi^{-1}(0) = \frac{1}{4} \begin{pmatrix} 5 & -1 \\ -1 & 1 \end{pmatrix}.$$

Note that

$$\Psi^{-1}(0)\mathbf{x}(0) = \frac{1}{4} \begin{pmatrix} 5 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}.$$

Therefore, the solution is

$$\mathbf{x}(t) = \Psi(t) \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} e^{-t} & e^{3t} \\ e^{-t} & 5e^{3t} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \boxed{\frac{1}{2} \begin{pmatrix} e^{-t} + e^{3t} \\ e^{-t} + 5e^{3t} \end{pmatrix}}.$$

As $t \rightarrow \infty$, the e^{3t} terms dominate since $|\lambda_2| > |\lambda_1|$, so $\|\mathbf{x}(t)\| \rightarrow \infty$. □

Question 12. Solve the given initial value problem. Describe the behavior of the solution as $t \rightarrow \infty$.

$$\mathbf{x}' = \begin{pmatrix} 0 & 0 & -1 \\ 2 & 0 & 0 \\ -1 & 2 & 4 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} 7 \\ 5 \\ 5 \end{pmatrix}$$

Solution. The characteristic equation is $\det(A - \lambda I) = 0$. Expanding along the first row:

$$-\lambda(\lambda^2 - 4\lambda) - (4 - \lambda) = -\lambda^3 + 4\lambda^2 + \lambda - 4 = 0.$$

$$\lambda^2(\lambda - 4) - (\lambda - 4) = 0 \implies (\lambda^2 - 1)(\lambda - 4) = 0 \implies \lambda_1 = -1, \quad \lambda_2 = 1, \quad \lambda_3 = 4.$$

For $\lambda_1 = -1$,

$$\begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ -1 & 2 & 5 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(1)} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \implies \mathbf{x}_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} e^{-t}.$$

For $\lambda_3 = 4$,

$$\begin{pmatrix} -4 & 0 & -1 \\ 2 & -4 & 0 \\ -1 & 2 & 0 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & \frac{1}{4} \\ 0 & 1 & \frac{1}{8} \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(3)} = \begin{pmatrix} 2 \\ 1 \\ -8 \end{pmatrix} \implies \mathbf{x}_3 = \begin{pmatrix} 2 \\ 1 \\ -8 \end{pmatrix} e^{4t}.$$

For $\lambda_2 = 1$,

$$\begin{pmatrix} -1 & 0 & -1 \\ 2 & -1 & 0 \\ -1 & 2 & 3 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(2)} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \implies \mathbf{x}_2 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} e^t.$$

The general solution is

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} e^t + c_3 \begin{pmatrix} 2 \\ 1 \\ -8 \end{pmatrix} e^{4t}.$$

Applying $\mathbf{x}(0) = (7, 5, 5)^T$:

$$\begin{pmatrix} 1 & 1 & 2 & 7 \\ -2 & 2 & 1 & 5 \\ 1 & -1 & -8 & 5 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -1 \end{pmatrix}.$$

So $c_1 = 3$, $c_2 = 6$, $c_3 = -1$. Therefore the solution is:

$$\mathbf{x}(t) = 6 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} e^t + 3 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} e^{-t} - \begin{pmatrix} 2 \\ 1 \\ -8 \end{pmatrix} e^{4t}.$$

As $t \rightarrow \infty$, the e^{4t} term dominates since $\lambda_3 = 4 > 0$, so $\|\mathbf{x}(t)\| \rightarrow \infty$. □

3 7.6

Question 6. Express the general solution of the given system of equations in terms of real-valued functions.

$$\mathbf{x}' = \begin{pmatrix} -3 & 0 & 2 \\ 1 & -1 & 0 \\ -2 & -1 & 0 \end{pmatrix} \mathbf{x}$$

Solution. The characteristic equation is $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} -3 - \lambda & 0 & 2 \\ 1 & -1 - \lambda & 0 \\ -2 & -1 & -\lambda \end{pmatrix} = 0.$$

Expanding along the first row,

$$(-3 - \lambda)(\lambda^2 + \lambda) + 2(-1 - 2 - 2\lambda) = 0 \implies \lambda^3 + 4\lambda^2 + 7\lambda + 6 = 0.$$

$$(\lambda + 2)(\lambda^2 + 2\lambda + 3) = 0 \implies \lambda_1 = -2, \quad \lambda_{2,3} = -1 \pm i\sqrt{2}.$$

For $\lambda_1 = -2$, we solve $(A + 2I)\mathbf{v} = \mathbf{0}$ so that

$$\begin{pmatrix} -1 & 0 & 2 \\ 1 & 1 & 0 \\ -2 & -1 & 2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(1)} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \implies \mathbf{x}_1 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} e^{-2t}.$$

For $\lambda_2 = -1 + i\sqrt{2}$, we solve $(A - \lambda_2 I)\mathbf{v} = \mathbf{0}$. Using row 2: $v_1 + (-i\sqrt{2})v_2 = 0 \implies v_1 = i\sqrt{2}v_2$. Setting $v_2 = 1$, then from row 1,

$$\begin{aligned} (-3 - (-1 + i\sqrt{2}))(i\sqrt{2}) + 2v_3 &= 0 \implies (-2 - i\sqrt{2})(i\sqrt{2}) + 2v_3 = 0 \\ \implies -2i\sqrt{2} + \sqrt{2} \cdot \sqrt{2} + 2v_3 &= 0 \implies 2 - 2i\sqrt{2} + 2v_3 = 0 \implies v_3 = -1 + i\sqrt{2}. \end{aligned}$$

So,

$$\mathbf{v} = \begin{pmatrix} i\sqrt{2} \\ 1 \\ -1 + i\sqrt{2} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + i \begin{pmatrix} \sqrt{2} \\ 0 \\ \sqrt{2} \end{pmatrix}.$$

The complex solution is

$$\begin{aligned} e^{(-1+i\sqrt{2})t} \left[\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + i \begin{pmatrix} \sqrt{2} \\ 0 \\ \sqrt{2} \end{pmatrix} \right] &= e^{-t} (\cos \sqrt{2}t + i \sin \sqrt{2}t) \left[\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + i \begin{pmatrix} \sqrt{2} \\ 0 \\ \sqrt{2} \end{pmatrix} \right] \\ &= e^{-t} \begin{pmatrix} -\sqrt{2} \sin \sqrt{2}t \\ \cos \sqrt{2}t \\ -\cos \sqrt{2}t - \sqrt{2} \sin \sqrt{2}t \end{pmatrix} + e^{-t} \begin{pmatrix} \sqrt{2} \cos \sqrt{2}t \\ \sin \sqrt{2}t \\ \sqrt{2} \cos \sqrt{2}t - \sin \sqrt{2}t \end{pmatrix} i. \end{aligned}$$

Therefore the general solution in terms of real-valued functions is:

$$\mathbf{x} = c_1 \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} e^{-2t} + c_2 e^{-t} \begin{pmatrix} -\sqrt{2} \sin \sqrt{2}t \\ \cos \sqrt{2}t \\ -\cos \sqrt{2}t - \sqrt{2} \sin \sqrt{2}t \end{pmatrix} + c_3 e^{-t} \begin{pmatrix} \sqrt{2} \cos \sqrt{2}t \\ \sin \sqrt{2}t \\ \sqrt{2} \cos \sqrt{2}t - \sin \sqrt{2}t \end{pmatrix}.$$

□

Question 8. Find the solution of the given initial value problem. Describe the behavior of the solution as $t \rightarrow \infty$.

$$\mathbf{x}' = \begin{pmatrix} -3 & 2 \\ -1 & -1 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Solution. The characteristic equation is

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{pmatrix} -3 - \lambda & 2 \\ -1 & -1 - \lambda \end{pmatrix} = (-3 - \lambda)(-1 - \lambda) + 2 = \lambda^2 + 4\lambda + 5 = 0 \\ \implies \lambda &= \frac{-4 \pm \sqrt{16 - 20}}{2} = -2 \pm i. \end{aligned}$$

So $\lambda_1 = -2 + i$ and $\lambda_2 = -2 - i$. For $\lambda_1 = -2 + i$,

$$\begin{pmatrix} -1 - i & 2 \\ -1 & 1 - i \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & -1 + i \\ 0 & 0 \end{pmatrix} \implies \mathbf{v} = \begin{pmatrix} 1 - i \\ 1 \end{pmatrix}.$$

The complex solution is

$$e^{(-2+i)t} \begin{pmatrix} 1 - i \\ 1 \end{pmatrix} = e^{-2t}(\cos t + i \sin t) \begin{pmatrix} 1 - i \\ 1 \end{pmatrix} = e^{-2t} \begin{pmatrix} (1 - i)(\cos t + i \sin t) \\ \cos t + i \sin t \end{pmatrix}.$$

Expanding $(1 - i)(\cos t + i \sin t)$:

$$= \cos t + i \sin t - i \cos t + \sin t = (\cos t + \sin t) + i(\sin t - \cos t).$$

Separating into real and imaginary parts $\mathbf{a} + i\mathbf{b}$:

$$\mathbf{a} = e^{-2t} \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix}, \quad \mathbf{b} = e^{-2t} \begin{pmatrix} \sin t - \cos t \\ \sin t \end{pmatrix}.$$

The general solution is:

$$\mathbf{x} = c_1 \mathbf{a} + c_2 \mathbf{b} = c_1 e^{-2t} \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} \sin t - \cos t \\ \sin t \end{pmatrix}.$$

Applying $\mathbf{x}(0) = (1, -2)^T$:

$$c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Row reducing the augmented matrix:

$$\begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & -2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \end{pmatrix} \implies c_1 = -2, \quad c_2 = -3.$$

Therefore the solution is:

$$\mathbf{x}(t) = -2e^{-2t} \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix} - 3e^{-2t} \begin{pmatrix} \sin t - \cos t \\ \sin t \end{pmatrix} = \boxed{e^{-2t} \begin{pmatrix} \cos t - 5 \sin t \\ -2 \cos t - 3 \sin t \end{pmatrix}}.$$

As $t \rightarrow \infty$, since $\text{Re}(\lambda) = -2 < 0$, the factor $e^{-2t} \rightarrow 0$ and therefore $\|\mathbf{x}(t)\| \rightarrow 0$. The solution decays to zero while oscillating. $\square$

Question 13. The coefficient matrix contains a parameter α .

- (a) Determine the eigenvalues in terms of α .
- (b) Find the bifurcation value or values of α where the qualitative nature of the phase portrait for the system changes.

$$\mathbf{x}' = \begin{pmatrix} \frac{5}{4} & \frac{3}{4} \\ \alpha & \frac{5}{4} \end{pmatrix} \mathbf{x}$$

Solution. For part (a), the characteristic equation is

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{pmatrix} \frac{5}{4} - \lambda & \frac{3}{4} \\ \alpha & \frac{5}{4} - \lambda \end{pmatrix} = \left(\frac{5}{4} - \lambda \right)^2 - \frac{3\alpha}{4} = 0 \\ \implies \lambda &= \frac{5}{4} \pm \sqrt{\frac{3\alpha}{4}} = \frac{5}{4} \pm \frac{\sqrt{3\alpha}}{2}. \end{aligned}$$

For part (b), the qualitative nature of the phase portrait changes when the eigenvalues transition between real and complex, or when their real part changes sign.

Bifurcation 1: The eigenvalues change from real to complex when the discriminant $\frac{3\alpha}{4} = 0$, i.e.:

$$\alpha = 0.$$

For $\alpha > 0$, there are two distinct real eigenvalues. For $\alpha < 0$, the complex conjugate eigenvalues have real part $\frac{5}{4} > 0$.

Bifurcation 2: When $\alpha < 0$, the eigenvalues are complex with real part $\frac{5}{4} \neq 0$, so no further bifurcation occurs from sign change of real part.

Bifurcation 3: When $\alpha > 0$, the eigenvalues are real. They change sign when

$$\frac{5}{4} - \frac{\sqrt{3\alpha}}{2} = 0 \implies \sqrt{3\alpha} = \frac{5}{2} \implies \alpha = \frac{25}{12}.$$

For $0 < \alpha < \frac{25}{12}$, both eigenvalues are positive. For $\alpha > \frac{25}{12}$: eigenvalues have opposite signs. Therefore, the bifurcation values are

$$\boxed{\alpha = 0 \quad \text{and} \quad \alpha = \frac{25}{12}.$$

□

4 7.7

Question 4.

- (a) Find a fundamental matrix for the given system of equations.
 (b) Find the fundamental matrix $\Phi(t)$ satisfying $\Phi(0) = \mathbf{I}$.

$$\mathbf{x}' = \begin{pmatrix} -1 & -4 \\ 1 & -1 \end{pmatrix} \mathbf{x}$$

Solution. The characteristic equation is

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{pmatrix} -1 - \lambda & -4 \\ 1 & -1 - \lambda \end{pmatrix} = (-1 - \lambda)^2 + 4 = \lambda^2 + 2\lambda + 5 = 0 \\ \implies \lambda &= \frac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i. \end{aligned}$$

So $\lambda_1 = -1 + 2i$ and $\lambda_2 = -1 - 2i$. For $\lambda_1 = -1 + 2i$,

$$\begin{pmatrix} -2i & -4 \\ 1 & -2i \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & -2i \\ 0 & 0 \end{pmatrix} \implies \mathbf{v} = \begin{pmatrix} 2i \\ 1 \end{pmatrix}.$$

The complex solution is

$$\begin{aligned} e^{(-1+2i)t} \begin{pmatrix} 2i \\ 1 \end{pmatrix} &= e^{-t}(\cos 2t + i \sin 2t) \begin{pmatrix} 2i \\ 1 \end{pmatrix} = e^{-t} \begin{pmatrix} 2i \cos 2t - 2 \sin 2t \\ \cos 2t + i \sin 2t \end{pmatrix} \\ &= e^{-t} \begin{pmatrix} -2 \sin 2t \\ \cos 2t \end{pmatrix} + i e^{-t} \begin{pmatrix} 2 \cos 2t \\ \sin 2t \end{pmatrix}. \end{aligned}$$

So the general solution is

$$\mathbf{x} = c_1 e^{-t} \begin{pmatrix} -2 \sin 2t \\ \cos 2t \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} 2 \cos 2t \\ \sin 2t \end{pmatrix}.$$

For part (a), the fundamental matrix is

$$\Psi(t) = e^{-t} \begin{pmatrix} -2 \sin 2t & 2 \cos 2t \\ \cos 2t & \sin 2t \end{pmatrix}.$$

For part (b), we need $\Phi(t) = \Psi(t)\Psi^{-1}(0)$. At $t = 0$,

$$\Psi(0) = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \implies \Psi^{-1}(0) = \frac{1}{-2} \begin{pmatrix} 0 & -2 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 0 \end{pmatrix}.$$

Therefore,

$$\Phi(t) = \Psi(t)\Psi^{-1}(0) = e^{-t} \begin{pmatrix} -2 \sin 2t & 2 \cos 2t \\ \cos 2t & \sin 2t \end{pmatrix} \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} = \boxed{e^{-t} \begin{pmatrix} \cos 2t & -2 \sin 2t \\ \frac{1}{2} \sin 2t & \cos 2t \end{pmatrix}}.$$

□

Question 7.

- (a) Find a fundamental matrix for the given system of equations.
(b) Find the fundamental matrix $\Phi(t)$ satisfying $\Phi(0) = \mathbf{I}$.

$$\mathbf{x}' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -8 & -5 & -3 \end{pmatrix} \mathbf{x}$$

Solution. The characteristic equation is $\det(A - \lambda I) = 0$. and Expanding simplifying gives:

$$-\lambda^3 - \lambda^2 + 4\lambda - \dots = 0.$$

Testing $\lambda = -2$: $(-2)^3 + (-2)^2 + \dots = 0$ ✓. Factoring:

$$(\lambda + 2)(\lambda^2 - \lambda + 2) = 0 \implies \lambda_1 = -2, \quad \lambda_{2,3} = \frac{1 \pm i\sqrt{7}}{2}.$$

For $\lambda_1 = -2$, we solve $(A + 2I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 3 & 1 & 1 \\ 2 & 3 & -1 \\ -8 & -5 & -1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & \frac{2}{7} \\ 0 & 1 & -\frac{5}{7} \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(1)} = \begin{pmatrix} -2 \\ 5 \\ 7 \end{pmatrix} \implies \mathbf{x}_1 = \begin{pmatrix} -2 \\ 5 \\ 7 \end{pmatrix} e^{-2t}.$$

For $\lambda_2 = \frac{1}{2} + \frac{\sqrt{7}}{2}i$, solving $(A - \lambda_2 I)\mathbf{v} = \mathbf{0}$ gives:

$$\mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + i \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

The complex solution is

$$e^{t/2} \left(\cos \frac{\sqrt{7}}{2}t + i \sin \frac{\sqrt{7}}{2}t \right) \left[\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + i \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right].$$

Separating into real and imaginary parts $\mathbf{a} + i\mathbf{b}$:

$$\mathbf{a} = e^{t/2} \begin{pmatrix} \sin \frac{\sqrt{7}}{2}t \\ \cos \frac{\sqrt{7}}{2}t \\ -\cos \frac{\sqrt{7}}{2}t - \sin \frac{\sqrt{7}}{2}t \end{pmatrix}, \quad \mathbf{b} = e^{t/2} \begin{pmatrix} -\cos \frac{\sqrt{7}}{2}t \\ \sin \frac{\sqrt{7}}{2}t \\ -\sin \frac{\sqrt{7}}{2}t + \cos \frac{\sqrt{7}}{2}t \end{pmatrix}.$$

For part (a), the fundamental matrix is:

$$\Psi(t) = \begin{pmatrix} -2e^{-2t} & e^{t/2} \sin \frac{\sqrt{7}}{2}t & -e^{t/2} \cos \frac{\sqrt{7}}{2}t \\ 5e^{-2t} & e^{t/2} \cos \frac{\sqrt{7}}{2}t & e^{t/2} \sin \frac{\sqrt{7}}{2}t \\ 7e^{-2t} & -e^{t/2} \left(\cos \frac{\sqrt{7}}{2}t + \sin \frac{\sqrt{7}}{2}t \right) & e^{t/2} \left(\cos \frac{\sqrt{7}}{2}t - \sin \frac{\sqrt{7}}{2}t \right) \end{pmatrix}.$$

□